Roll no____________________ Section:__________________

```
float calculateBill(int units, bool isIndustrial, bool hasArrears) {
    float rate = 0;
    float total = 0;
    if (units <= 100) {
        rate = 5.0;
    } else {
        if (units <= 300) { // units > 100 is implied by the 'else'
            rate = 7.5;
        } else {
            rate = 10.0;
        }
    }
    if (isIndustrial) {
        if (units > 200) {
            rate = rate * 0.8;
        }
    }
    total = units * rate;
    if (hasArrears) {
        total = total + (total * 0.1);
    }
    return total;
}
```

Part 1: Control Flow Testing (4 + 6 Points)

1.1. Construct the Graph: Draw the **Control Flow Graph (CFG)** for the `calculateBill` function. Ensure each node corresponds to a specific line or block of code and all decision branches (True/False) are clearly labeled.

1.2. Design Test Cases: Provide the **minimum** number of test cases required to achieve the following coverage criteria:

- **A. Statement Coverage, B. Branch Coverage, C. Condition Coverage**

Part 2: Data Flow Testing (4+3+3 Points)

Provide test cases to satisfy the following Data Flow criteria for the entire program:

- **A. All-Definitions Path, B. All-Uses Path, C. All-DU Paths**